

HS 529: Natural Resource and Development

Course Code	HS 529
Course Name	Natural Resource and Development
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	Post graduate students in SHSS
Distribution	Core course for M.A. in Development Studies
Approval	24th BoA (Apr 2018); 16th Senate (Apr 2018)

Course Contents

- **Natural resources and their role in sustainable energy** [14 Lectures]
 - Introduction: Natural Resources renewable v/s non-renewable. What is renewable energy? Why do we need renewable energy? Different renewable energy sources. (3 hours)
 - Solar energy: potential of solar energy reaching earth surface, collecting sunlight, solar photovoltaic and solar thermal techniques, solar power plants, solar water desalination, solar dryers, future challenges (3 hours)
 - Wind energy: Wind energy availability and basic working principle of wind turbines, resource assessment overview, modern wind turbines, installations and wind farms, advantages and limitations of wind farms (3 hours)
 - Biomass energy: Organic matters available on renewable basis like forests, agricultural, mill and industrial wastes etc., direct fired plants, co fired power plants, gasification (3 lectures)
 - Limitations of renewable sources. (2 hours)
- **Sustainable Development and Pollution Prevention** [14 Lectures]
 - Natural Cycles: Water cycle, oxygen cycle, phosphorous cycle, Appreciation of disturbance in these cycles as cause of pollution. (2 lectures)
 - Water Pollution Control and Technologies: Indian potable water and treated waste water standards, Health impact of potable drinking water, sources of water pollutants for urban and rural habitats, Waste water treatment technologies (4 lectures)
 - Air Pollution Control Technologies: Gaseous and particulate emissions, Health impact, sources of gaseous pollutants for urban and rural habitats, Air pollution control technologies
 - Solid Waste Management: Case Studies on Best Practices of Domestic Solid Waste Management/ Case Studies on Electronic Waste Management Practices (4 lectures)
- **Introduction to Biodiversity and Conservation** [14 Lectures]
 - The key concepts of biodiversity, its definition, meaning and functional significance (2 lectures)
 - Conservation of natural resources - Water, Soil, Flora, Fauna, endangered species etc.; In-situ and Ex-situ conservation (2 lectures)
 - International and National conventions (2 lectures)
 - Biodiversity and Conservation for Development - Poverty alleviation (2 lectures)
 - Critical approaches to Biodiversity conservation - policy and practice (1 lecture)
 - Biodiversity Technologies for Conservation (2 lectures)
 - Selected Case studies (3 lectures)

Text Books

1. Ruth Weiner Robin Matthews, **Environmental Engineering**, 4th Edition, Elsevier, 2003.
2. Aldo V.da Rosa, **Fundamental of Renewable Energy Processes**, Elsevier Press, 2009.
3. Michael J Jeffriesby, **Biodiversity and Conservation** (Routledge Introductions to Environment series), 2nd Edition, 2006.

4. References

1. Gautam, A., De, S., Dhar, A., Gupta, J.G., Pandey, A. (Eds.), **Sustainable Energy and Transportation**, Springer Singapore, 2018, ISBN 978-981-10-7508-7
2. <http://worldhappiness.report/ed/2017/>
3. <http://www.epa.ohio.gov/ocapp/p2/sustainable.aspx#124473580-practices>
4. Jahangir Hossain, Mahmud Apel, **Large Scale Renewable Power Generation: Advances in Technologies for Generation, Transmission and Storage** (Green Energy and Technology), Springer, 2014.